

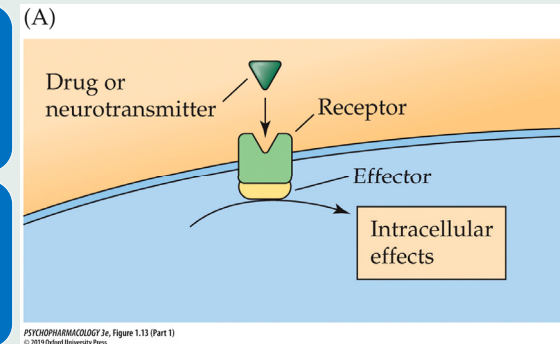
PHARMACODYNAMICS

- Interactions between the drug and its target (receptor); mechanisms of drug action
- Concentration-response relationships
- Receptor affinity
- Drug potency & efficacy

WHAT ABOUT RECEPTORS?

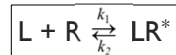
What is a receptor?

What do they do?



LIGAND-RECEPTOR INTERACTION

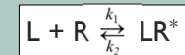
is dynamic and reversible



is in a constant state of binding (association) and dissociation (breaking away)

LIGAND-RECEPTOR INTERACTION

follows the law of mass action



- [L]igand combines with [R]eceptor to form an activated ligand-receptor complex [LR*]
- at equilibrium, the rate of LR* formation is equal to the rate of L + R dissociation

$$[L] [R] k_1 = [LR^*] k_2$$

The concentration of [L] determines the biological effect

RECEPTOR OCCUPANCY DEPENDS ON AFFINITY OF THE RECEPTOR FOR LIGAND

L-R interactions can be described by an affinity constant

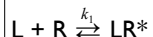
$$K_{\text{AFF}} = k_1 / k_2$$

or more commonly by a dissociation constant

$$K_d = k_2 / k_1$$

- units of K_d are M (mM, μ M, nM)
- lower K_d represents higher affinity

Law of mass action

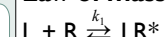


$$[L] [R] k_1 = [LR^*] k_2$$

RECEPTOR OCCUPANCY DEPENDS ON AFFINITY OF THE RECEPTOR FOR LIGAND

$[B_{\text{max}}]$ = total # of binding sites (receptors)

Law of mass action



$$[L] [R] k_1 = [LR^*] k_2$$

$[B_{\text{max}}]$ = bound receptors + free (unbound) receptors

$$[B_{\text{max}}] = [LR^*] + [R]$$

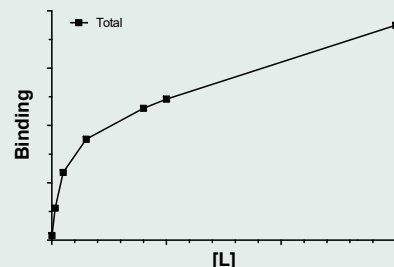
Fractional (or %) occupancy

then $K_d = \frac{[L] ([B_{\text{max}}] - [LR^*])}{[LR^*]}$, and

$$\frac{[LR^*]}{[B_{\text{max}}]} = \frac{[L]}{[L] + [K_d]}$$

RADIOLIGAND BINDING STUDIES

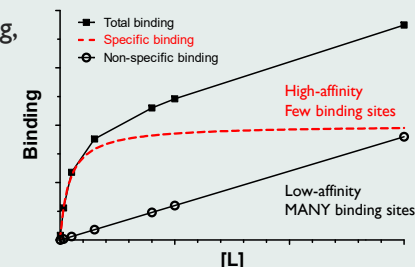
- A radioligand is a ligand that has been labeled with a radioactive element (e.g., ^3H or ^{125}I).
- Radioligand is incubated with a tissue preparation that contains the binding sites of interest.
- Establish equilibrium (30 – 60 min).



Total binding increases with higher concentration of radioligand [L].

TOTAL BINDING INCLUDES BOTH SPECIFIC AND NON-SPECIFIC BINDING

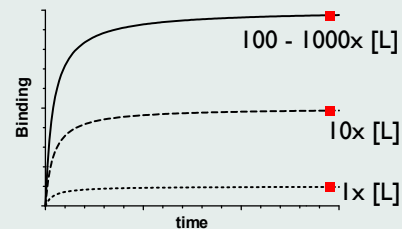
- To determine non-specific binding, an excess (very high conc.) of unlabeled ligand is added.
- Excess unlabeled (“cold”) ligand outcompetes (“hot”) radioligand for specific binding sites



Receptor binding is specific (---), high-affinity, and saturable.

EQUILIBRIUM BINDING CURVES ARE USED TO DETERMINE B_{MAX}

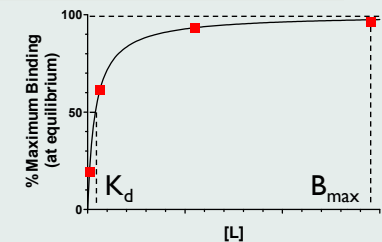
- Specific binding is determined for several concentrations of radioligand [L].
 - at equilibrium
 - (note: X-axis is time)
- At very high concentrations of [L], all (100%) of specific binding sites are occupied with radioligand



Equilibrium binding curves show *saturation* of specific binding at many [L]

EQUILIBRIUM BINDING AT SEVERAL [L] REVEALS K_D AND B_{MAX}

- Specific binding at equilibrium is plotted for several concentrations of radioligand [L] (note: X-axis is concentration)
- At the K_d concentration of [L], 50% of the maximum binding sites (B_{max}) are occupied with radioligand.

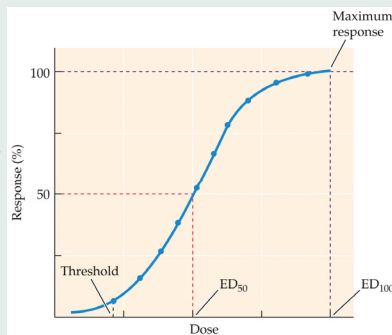


This is called a *saturation curve*

(Technically, both K_d & B_{max} are concentrations)

LIGAND – RECEPTOR INTERACTION

- can be described by a *concentration-response (dose-response)* relationship
- depicts the magnitude of drug effects at progressively larger doses
- The response can be
 - pharmacological (receptor binding)
 - biochemical (receptor activation)
 - physiological (neural activity)
 - behavioral (whole animal)



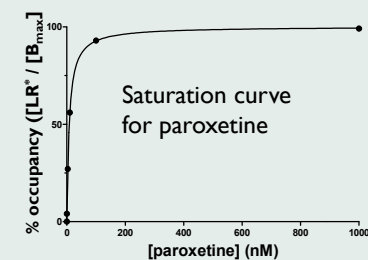
PSYCHOPHARMACOLOGY 4e, Figure 1.17
© 2012 Oxford University Press

LET'S MAKE SOME SIMPLE CALCULATIONS

- The antidepressant paroxetine (Paxil) blocks the 5-HT transporter with a $K_d \approx 8$ nM

$$\frac{[LR^*]}{[B_{max}]} = \frac{[L]}{[L] + [K_d]}$$

[paroxetine] [L]	% occupancy $\frac{[LR^*]}{[B_{max}]}$
0.3 nM	4%
3 nM	27%
10 nM	56%
100 nM	93%

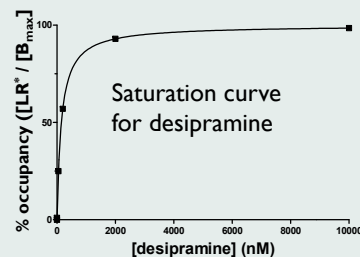


LET'S MAKE SOME SIMPLE CALCULATIONS

- The AD desipramine (Norpramin) blocks the 5-HT transporter with a $K_d \approx 150$ nM

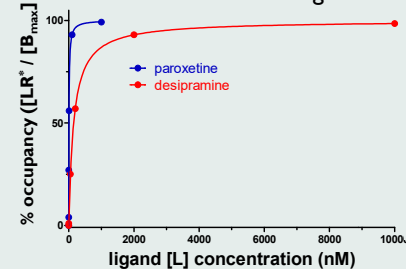
$$\frac{[LR^*]}{[B_{max}]} = \frac{[L]}{[L] + [K_d]}$$

[desipramine] [L]	% occupancy [LR*] / [B _{max}]
2 nM	1%
50 nM	25%
200 nM	57%
2000 nM	93%

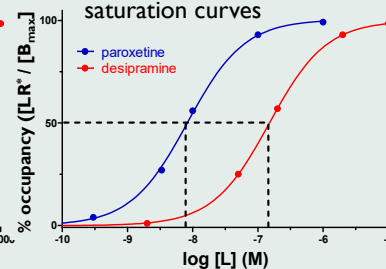


K_D IS THE CONCENTRATION WHERE 50% OF RECEPTORS ARE OCCUPIED

Saturation curves for 2 ligands

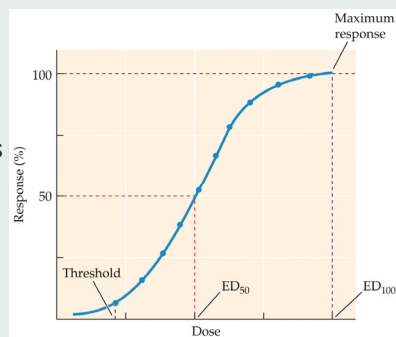


Semi-logarithmic plot of saturation curves



LIGAND – RECEPTOR INTERACTION

- can be described by a *concentration-response (dose-response)* relationship
- depicts the magnitude of drug effects at progressively larger doses
- The response can be
 - pharmacological (receptor binding)
 - biochemical (receptor activation)
 - physiological (neural activity)
 - behavioral (whole animal)



LIGANDS CAN HAVE DIFFERING PHARMACOLOGICAL ACTIONS

Drugs that *mimic* the effect of the endogenous transmitter

Agonist
Drug action/effect

Drugs that cannot produce the full agonist response

Partial agonist
Intermediate drug action/effect

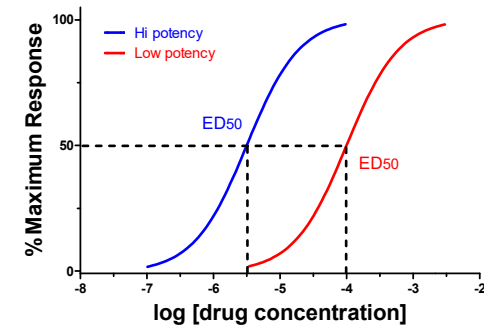
Drugs that *block* the effect of the endogenous transmitter

Antagonist
No drug action/effect

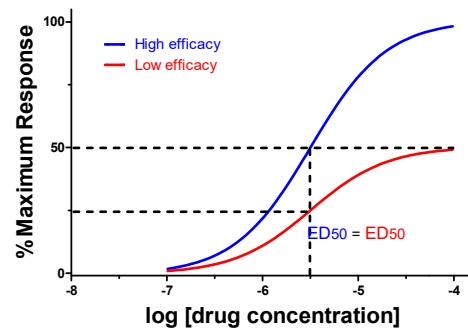
DRUG-RECEPTOR INTERACTIONS

- | | |
|-------------------|---|
| Receptor affinity | <ul style="list-style-type: none">• how tightly the receptor holds onto the drug• lock & key mechanism |
| Drug potency | <ul style="list-style-type: none">• the amount (concentration) of drug necessary to produce a desired level of effect |
| Drug efficacy | <ul style="list-style-type: none">• the maximum response produced by a drug. |

DRUG-RECEPTOR INTERACTIONS



DRUG-RECEPTOR INTERACTIONS



ANTAGONIST ACTIONS MAY BE COMPETITIVE OR NON-COMPETITIVE

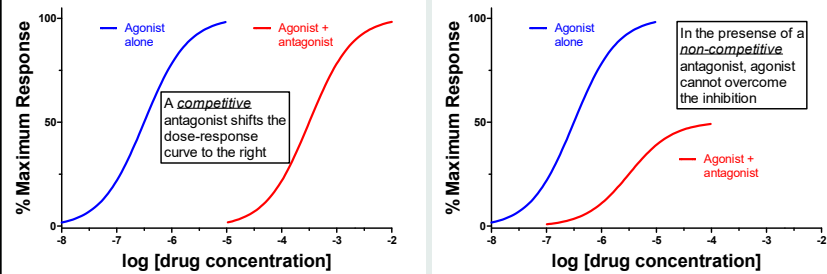
Competitive antagonists “compete” with the agonist for binding to the agonist binding site

- high affinity for the agonist binding site, but no efficacy

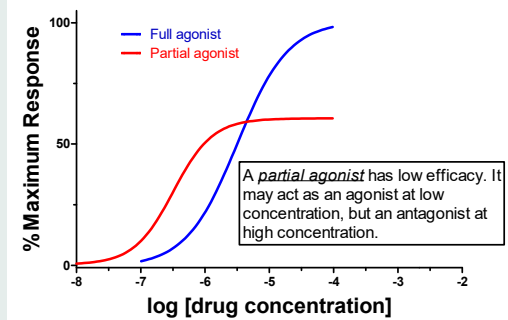
Non-competitive antagonists bind to a site different from the agonist binding site.

- negative allosteric modulation

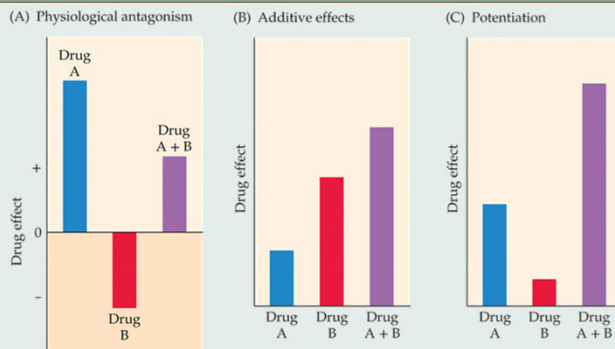
ANTAGONIST ACTIONS MAY BE COMPETITIVE OR NON-COMPETITIVE



PARTIAL AGONISTS HAVE LOW EFFICACY



BIOBEHAVIORAL INTERACTIONS CAN RESULT IN DIFFERENT DRUG EFFECTS



PSYCHOPHARMACOLOGY 4e, Figure 1.21
© 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins